1 Polynomial Functors

Give semantics to W-Types, which should be enough for our case (hopefully). By encoding a set of operations an their arities, which we can think of as the type constructors. Where B are the operations and A the arities, we each also fibre them over a set of types. We only consider endofunctors here.

$$
\begin{tikzcd}
 & A \arrow[r] \arrow[dl] & B \arrow[dr] & \\
I & & & I
\end{tikzcd}
$$

2 Given Constructors

We need the subtype to be stable under these, so we need an $F - Alg$ for the polynomial functor build from the constructors.

$$
\begin{tikzcd}
F(\Sigma_{x:X} P(x)) \arrow[d, "\alpha"] \\
\Sigma_{x:X} P(x)
\end{tikzcd}
$$

Therefore, we need something like $F(P)(F(x)) \implies P(x)$, from now on called a predicate being stable under F. Further if P is the smallest such predicate, so for all predicates Q stable under F we have $P(x) \implies Q(x)$ we can then $\Sigma_{x:X} P(x)$ should be initial under the Σ types, by just applying F to the obvious function. Why this is unique needs some work still.

$$
\begin{tikzcd}
F(\Sigma_{x:X} P(x)) \arrow[r, "F(f)"] \arrow[d, "\alpha"] & F(\Sigma_{x:X} Q(x)) \arrow[d, "\delta"] \\
\Sigma_{x:X} P(x) \arrow[r, "f"] & \Sigma_{x:X} Q(x)
\end{tikzcd}
$$

Finally, by building a prediacte which should be something like $\forall a : A \exists a' : A \;\; s.t. \;\; a = \tau(F(a'))$ we where $F(A) \rightarrow_{\tau} A$ is some F algebra. We should be able to build a subtype with an F-stable predicate that is isomorphic to A and therefore get initiality for all algebras.